iPhone Sales Analytics Platform

Name: Binayak Shrestha
Course: Data Engineering Bootcamp
Date: March 19th, 2026

Introduction

This project implements a data pipeline using Hive and HDFS following the Medallion Architecture. The data is processed through Bronze, Silver, and Gold layers to clean, transform, and analyze iPhone sales data. The final layer provides meaningful business insights such as revenue by product and store.

Database and Tables

SHOW DATABASES;
USE iphone_analytics;
SHOW TABLES;

```
hive (default)> SHOW DATABASES;
OK
default
hive_test
iphone_analytics
practice
xyz
Time taken: 0.307 seconds, Fetched: 5 row(s)
hive (default)> USE iphone_analytics;
OK
Time taken: 0.022 seconds
hive (iphone_analytics)> SHOW TABLES;
OK
bronze_customers
bronze_products
bronze_sales
bronze_stores
dim_customer
dim_date
dim_product
dim_store
fact_sales
raw_customers
raw_products
raw_sales
raw_stores
silver_customers
silver_products
silver_sales
silver_stores
```

Bronze Layer

```
hive (iphone_analytics)> SHOW TABLES;
OK
bronze_customers
bronze_products
bronze_sales
bronze_stores
```

Silver Layer

DESCRIBE FORMATTED silver_sales;

SHOW PARTITIONS silver_sales;

```
hive (iphone_analytics)> DESCRIBE FORMATTED silver_sales;
OK
# col_name              data_type               comment
sale_id                 int
product_id              int
customer_id             int
store_id                int
quantity                int

# Partition Information
# col_name              data_type               comment
sale_date               date

# Detailed Table Information
Database:               iphone_analytics
OwnerType:              USER
Owner:                  takeo
CreateTime:             Fri Mar 20 00:43:02 IST 2026
LastAccessTime:         UNKNOWN
Retention:              0
Location:               hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/silver_sales
Table Type:             MANAGED_TABLE
Table Parameters:
        COLUMN_STATS_ACCURATE   {\"BASIC_STATS\":\"true\"}
        bucketing_version       2
        numFiles                4
        numPartitions           4
        numRows                 4
        rawDataSize             20
        totalSize               3548
        transient_lastDdlTime   1773947582
```

```
# Storage Information
SerDe Library:          org.apache.hadoop.hive.ql.io.parquet.serde.ParquetHiveSerDe
InputFormat:            org.apache.hadoop.hive.ql.io.parquet.MapredParquetInputFormat
OutputFormat:           org.apache.hadoop.hive.ql.io.parquet.MapredParquetOutputFormat
Compressed:             No
Num Buckets:            -1
Bucket Columns:         []
Sort Columns:           []
Storage Desc Params:
        serialization.format    1
Time taken: 0.399 seconds, Fetched: 40 row(s)
```

Gold Layer

DESCRIBE FORMATTED fact_sales;

SHOW PARTITIONS fact_sales;

```
hive (iphone_analytics)> DESCRIBE FORMATTED fact_sales;
OK
# col_name              data_type               comment
sale_id                 int
customer_id             int
product_id              int
store_id                int
quantity                int
total_amount            int

# Partition Information
# col_name              data_type               comment
date_key                date

# Detailed Table Information
Database:               iphone_analytics
OwnerType:              USER
Owner:                  takeo
CreateTime:             Fri Mar 20 00:45:02 IST 2026
LastAccessTime:         UNKNOWN
Retention:              0
Location:               hdfs://0.0.0.0:54310/data/hive/warehouse/iphone_analytics.db/fact_sales
Table Type:             MANAGED_TABLE
Table Parameters:
        COLUMN_STATS_ACCURATE   {\"BASIC_STATS\":\"true\"}
        bucketing_version       2
        numFiles                4
        numPartitions           4
        numRows                 4
        rawDataSize             24
        totalSize               4160
        transient_lastDdlTime   1773947702
```

```
# Storage Information
SerDe Library:          org.apache.hadoop.hive.ql.io.parquet.serde.ParquetHiveSerDe
InputFormat:            org.apache.hadoop.hive.ql.io.parquet.MapredParquetInputFormat
OutputFormat:           org.apache.hadoop.hive.ql.io.parquet.MapredParquetOutputFormat
Compressed:             No
Num Buckets:            -1
Bucket Columns:         []
Sort Columns:           []
Storage Desc Params:
        serialization.format    1
Time taken: 0.083 seconds, Fetched: 41 row(s)
hive (iphone_analytics)> SHOW PARTITIONS fact_sales;
OK
date_key=2023-01-10
date_key=2023-01-12
date_key=2023-01-15
date_key=2023-01-20
Time taken: 0.046 seconds, Fetched: 4 row(s)
```

HDFS Proof

/data/hive/warehouse/iphone_analytics.db

```
takeo@1eef5da3469c:~$ hadoop fs -fs hdfs://localhost:54310 -ls /data/hive/warehouse/iphone_analytics.db
2026-03-20 02:01:46,098 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applic
able
Found 17 items
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:41 /data/hive/warehouse/iphone_analytics.db/bronze_customers
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:41 /data/hive/warehouse/iphone_analytics.db/bronze_products
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:42 /data/hive/warehouse/iphone_analytics.db/bronze_sales
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:42 /data/hive/warehouse/iphone_analytics.db/bronze_stores
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:45 /data/hive/warehouse/iphone_analytics.db/dim_customer
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:46 /data/hive/warehouse/iphone_analytics.db/dim_date
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:45 /data/hive/warehouse/iphone_analytics.db/dim_product
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:45 /data/hive/warehouse/iphone_analytics.db/dim_store
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:47 /data/hive/warehouse/iphone_analytics.db/fact_sales
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:40 /data/hive/warehouse/iphone_analytics.db/raw_customers
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:40 /data/hive/warehouse/iphone_analytics.db/raw_products
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:40 /data/hive/warehouse/iphone_analytics.db/raw_sales
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:40 /data/hive/warehouse/iphone_analytics.db/raw_stores
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:43 /data/hive/warehouse/iphone_analytics.db/silver_customers
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:44 /data/hive/warehouse/iphone_analytics.db/silver_products
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:44 /data/hive/warehouse/iphone_analytics.db/silver_sales
drwxr-xr-x   - takeo supergroup          0 2026-03-20 00:44 /data/hive/warehouse/iphone_analytics.db/silver_stores
takeo@1eef5da3469c:~$
```

Row Counts

SELECT COUNT(*) FROM bronze_sales;
SELECT COUNT(*) FROM silver_sales;
SELECT COUNT(*) FROM fact_sales;

```
hive (iphone_analytics)> SHOW PARTITIONS fact_sales;
OK
date_key=2023-01-10
date_key=2023-01-12
date_key=2023-01-15
date_key=2023-01-20
Time taken: 0.046 seconds, Fetched: 4 row(s)
hive (iphone_analytics)> USE iphone_analytics;
OK
Time taken: 0.038 seconds
hive (iphone_analytics)> SELECT COUNT(*) FROM bronze_sales;
OK
4
Time taken: 0.853 seconds, Fetched: 1 row(s)
hive (iphone_analytics)> SELECT COUNT(*) FROM silver_sales;
OK
4
Time taken: 0.083 seconds, Fetched: 1 row(s)
hive (iphone_analytics)> SELECT COUNT(*) FROM fact_sales;
OK
4
Time taken: 0.082 seconds, Fetched: 1 row(s)
hive (iphone_analytics)>
```

Sample Data

SELECT * FROM silver_sales LIMIT 10;
SELECT * FROM fact_sales LIMIT 10;

```
hive (iphone_analytics)> SELECT * FROM silver_sales LIMIT 10;
OK
1       101     1       10      1       2023-01-10
2       102     2       20      2       2023-01-12
3       101     3       20      1       2023-01-15
4       103     1       10      2       2023-01-20
Time taken: 0.094 seconds, Fetched: 4 row(s)
hive (iphone_analytics)> SELECT * FROM fact_sales LIMIT 10;
OK
1       1       101     10      1       900     2023-01-10
2       2       102     20      2       2200    2023-01-12
3       3       101     20      1       900     2023-01-15
4       1       103     10      2       400     2023-01-20
Time taken: 0.086 seconds, Fetched: 4 row(s)
hive (iphone_analytics)>
```

Business Queries

Revenue by product

> SELECT p.product_name, SUM(f.total_amount) AS revenue

> FROM fact_sales f

> JOIN dim_product p

> ON f.product_id = p.product_id

> GROUP BY p.product_name;

```
Total MapReduce CPU Time Spent: 2 seconds 680 msec
OK
AirPods 400
iPhone 13       1800
iPhone 14       2200
Time taken: 18.654 seconds, Fetched: 3 row(s)
hive (iphone_analytics)>
```

Revenue by store

> SELECT s.store_name, SUM(f.total_amount) AS revenue

> FROM fact_sales f

> JOIN dim_store s

> ON f.store_id = s.store_id

> GROUP BY s.store_name;

```
Total MapReduce CPU Time Spent: 2 seconds 700 msec
OK
Apple Store NY  3100
Apple Store SJ  1300
Time taken: 15.854 seconds, Fetched: 2 row(s)
hive (iphone_analytics)>
```

Conclusion

This project successfully implemented a Hive-based data pipeline using Bronze, Silver, and Gold layers. Raw iPhone sales data was cleaned, transformed, and stored in structured tables for analysis. The final results provided useful business insights such as revenue by product and revenue by store.